

Future Proves Past

Using Post-NCAA Careers Abroad to Forecast Incoming International Recruits

Abstract. We forecast international players’ first and later NCAA seasons, conditional on a supplied destination team. Scarce direct examples are augmented by later international histories paired with observed NCAA outcomes. Shared components combine competition context, exposure- and recency-weighted game summaries, a separate season-summary channel, and age, time, and destination features. Opponent-adjusted ratings enter when available. A structured distribution generates playing opportunity and production, with an observation model for recorded minutes. We define historical information limits and a forward evaluation measuring the incremental gain from reconstruction supervision. Attention and other extensions remain alternatives. This paper specifies a method and evaluation protocol; it reports no measured NCAA transfer gain.

1 Introduction

A recruiting forecast must predict both playing opportunity and production. Competition, age, role, and destination context affect both. We predict a distribution over the player’s NCAA season, treating team membership as a supplied scenario.

The central opportunity is an imbalance in the data. Relatively few observed players move from international basketball into the NCAA, while many NCAA players later play internationally. Both trajectories connect international performance to a known NCAA season:

Task	International history used	NCAA outcome to predict
Forward	Games around age 19	Season opening at age 20; $\Delta \approx -1$.
Reconstruction	Games around age 23	Earlier season opening at age 20; $\Delta \approx +3$.

These are illustrative ages. The reconstruction player need not have a pre-NCAA history. Their later games and observed NCAA outcome form a complete auxiliary example. Only the forward task is used at deployment.

Our hypothesis is that shared game representations and outcome relationships can be learned more accurately from both samples. Age and signed time let the model distinguish the tasks; they do not make reconstruction an algebraic inverse or remove selection differences. The hypothesis succeeds only if adding reconstruction examples improves forecasts for future NCAA entrants.

1.1 Related work

Multitask learning supplies the shared-parameter principle [1]. Glazer’s G League-to-NBA translation factors [6] provide a close basketball comparison: matching and difference-in-differences estimate league effects on comparable statistics, subject to overlap and parallel-trends assumptions. Our target also includes future playing opportunity and development.

Twin Networks, Seer Forcing, and generalized distillation [2, 3, 7] show ways to use richer training-time information. Their paired-view mechanisms are extensions here; they do not directly supply absent pre-NCAA inputs. Back-translation [5] uses synthetic source examples, whereas our core task uses observed NCAA outcomes. Latent-state and adaptive-weighting alternatives [4, 8, 9] appear in the appendices.

2 Data and probability model

2.1 Prediction unit and records

The unit is player i on NCAA team $j = j(i, t)$ in season t . Choose one forecast horizon, such as the regular season; both the outcome aggregation and schedule length S_{jt} use that same horizon. If postseason or cancellations make the number of games uncertain, supply a forecast distribution for S_{jt} .

We use four record types: player game logs and season summaries, NCAA roster-season outcomes, dated player attributes, and team games/context. Game logs contain production, minutes, dates, teams, competition, and venue; the rating model adds pregame opponent strength. Attributes include date of birth, height, position, and eligibility/injury information available at the forecast date. Source competitions are explicitly listed international leagues or age tiers; other non-NCAA competitions require separate labels.

Positive minutes, nonzero player box-score counts, or a mapped provider participation status provide evidence of an appearance. Conflicting evidence remains unresolved pending audit; rounded zero minutes alone do not establish nonparticipation. A zero-game outcome requires evidence of nonparticipation over the full horizon, from reconciled records or provider season participation status. Primary likelihood units require complete, verified outcomes for both participants and nonparticipants. Non-rostered players receive no artificial zero label; input masks distinguish absent inputs from measured zeros.

Notation	Meaning
$t; T_t, E_t$	NCAA season index; its opening and ending dates.
$c_t; c^{\text{tr}}$	Forecast date for season t ; date the model is fit.
$w; \mathcal{H}_{iw}$	Source window: games and admissible season summaries.
$s_{iw}; \Delta_{iwt}$	Latest source date/summary endpoint; relative time to T_t .
$\mathcal{S}_{iw}; \mathcal{S}_{iw}^{\text{cov}}$	Encoded summaries; summaries used only for coverage.
s_{iw}^G	Latest date in the game channel.
$Z_{it}; C_{jt}^{(c_t)}$	Player attributes/experience; supplied team's available context.
$\mathbf{Y}_{it} = (G, J, M, \mathbf{B})$	Games, starts, recorded minutes, primitive box-score totals.
$M^*; \mathcal{P}$	Latent playing minutes; common pretraining pool.
$\mathbf{h}_{iw}; u$	Encoded history; final player-season representation.
$\mathcal{D}_-, \mathcal{D}_+; \Theta$	Forward/reconstruction samples; fitted network parameters.

2.2 What each direction may observe

Seasons use end-year labels assigned per competition: a record's season comes from a verified provider competition-season identifier or, failing that, from its competition's documented period boundaries; only where neither exists does an audited fallback apply, under which a game dated on or after 1 August belongs to the following labelled year. The original label, the assigned season, and the mapping version are retained with each record. This indexing convention is separate from the NCAA opening and ending dates T_t, E_t and from the actual periods covered by summaries; chronological eligibility depends on actual dates and release dates, never on labels. Represent dates in years for elapsed-time calculations. A nonempty window contains international games, admitted season summaries, or both. Its reference date s_{iw} is the latest included game date or summary endpoint, including summaries used only to assess game coverage. Summary periods must lie wholly on the appropriate side of the target season and satisfy release cutoffs. With date of birth dob_i , define

$$\Delta_{iwt} = s_{iw} - T_t, \quad a_i(s) = s - \text{dob}_i, \quad a_{it} = a_i(T_t) = a_i(s_{iw}) - \Delta_{iwt}. \quad (1)$$

Thus $\Delta < 0$ means the source precedes the NCAA target, and $\Delta > 0$ means it follows it. Reconstruction windows must begin after the NCAA season ends, not merely after its opening.

For preseason forecasts, $c_t \leq T_t$. Let $\text{rel}(x)$ be the release date of record x . The two cutoffs serve different purposes:

Input or label	Forward example	Reconstruction example
International games	Before T_t ; released by c_t .	After E_t ; released by c^{tr} .
Target attributes, prior NCAA games, team context	Frozen at c_t .	Also frozen at c_t .
NCAA label used in fitting	Released by c^{tr} .	Released by c^{tr} .

All fitted preprocessing and training records must also exist by c^{tr} . At deployment $c^{\text{tr}} \leq c_t$. Release dates are observed or assigned under a declared provider-lag assumption. Appendix D.2 collects the implementation defaults.

The default destination context is the previous season’s offensive rating, defensive rating, and pace, with availability masks:

$$C_{jt}^{(c_t)} = (\text{ORtg}_{j,t-1}, \text{DRtg}_{j,t-1}, \text{Pace}_{j,t-1}, \text{masks}). \quad (2)$$

The input X_{iwt} combines $\mathcal{H}_{iw}$, prior NCAA history $\mathcal{H}_{it}^{\text{N}}$, Z_{it} , $C_{jt}^{(c_t)}$, and Δ_{iwt} . Prior NCAA history is a separate channel and is empty for first-year entrants. Z_{it} includes target age and completed NCAA seasons. Both tasks model $p_{d,\Theta}(\mathbf{Y}_{it} \mid X_{iwt}, S_{jt})$, $d \in \{-, +\}$, with shared networks and direction-aware inputs. Their conditional distributions may differ.

2.3 Training samples and coverage

Each admissible triple (i, w, t) pairs a known NCAA outcome with an international window. Nonempty earlier windows form $\mathcal{D}_-$; nonempty later windows form $\mathcal{D}_+$. A player with both can contribute to both tasks. NCAA-only history-to-future-season examples form a separate optional sample $\mathcal{D}_{\text{N}}$ and are not counted again in $\mathcal{D}_-$.

Arms B, C, and C^{shuf} pretrain on the same fold-admissible international pool $\mathcal{P}$, including reconstruction histories with NCAA correspondence withheld. Exclude evaluation players and NCAA target-derived inputs from this pool. Repeated windows are dependent views, not independent players. Report unique players, target seasons, windows, coverage, and eligible reconstruction pairs by cohort and training cutoff. Appendix A defines the windows and source-only pretraining inputs.

2.4 A season distribution: opportunity before production

The model first generates opportunity and latent playing minutes M^* , then production. Recorded minutes M follow the provider’s reporting convention. Let O count overtime periods in games in which the player appears. Given representation u , the latent joint distribution is

$$p(g, j, o, m^*, \mathbf{b} \mid u, S) = p(g \mid u, S) p(j \mid g, u) p(o \mid g, u) \times p(m^* \mid g, j, o, u) p(\mathbf{b} \mid m^*, g, u). \quad (3)$$

The distribution for $\mathbf{Y} = (G, J, M, \mathbf{B})$ integrates a normalized recording kernel over M^* and sums out O (Appendix A.3). When $G = 0$, starts, latent and recorded minutes, and production are zero. A brief appearance may instead have $G > 0$ and recorded $M = 0$.

Quantity	Distribution	Why it is structured this way
Games G	Hurdle beta-binomial on $0:S$	Separates no participation from positive appearances.
Starts $J \mid G$	Beta-binomial on $0:G$	Starts cannot exceed games.
Latent minutes M^*	Beta fraction of available minutes, plus an atom at the maximum	Bounds exposure while allowing full participation.
Production $\mathbf{B}$	Ordered counts; makes conditional on attempts	Preserves nonnegative counts and scoring identities.

For $S \geq 1$, if $b(g; S, \alpha_G, \beta_G)$ is the beta-binomial mass function,

$$\Pr(G = 0) = 1 - \pi_G, \quad \Pr(G = g) = \pi_G \frac{b(g; S, \alpha_G, \beta_G)}{1 - b(0; S, \alpha_G, \beta_G)}, \quad 1 \leq g \leq S. \quad (4)$$

If $S = 0$, all season outcomes are deterministically zero. Probabilities use logistic links and positive parameters use softplus links. Appendix A.2 gives the minutes distribution, overtime treatment, rounding convention, and complete count order.

2.5 Points, efficiency, and PER

For shot type $r \in \{2, 3, f\}$, let A_r and K_r denote attempts and makes. The count heads use

$$\begin{aligned} A_r \mid M^*, u, \text{earlier counts} &\sim \text{NegBin}((M^*/L)\lambda_r, \kappa_r), \\ K_r \mid A_r, M^*, u, \text{earlier counts} &\sim \text{BetaBin}(A_r, q_r\tau_r, (1 - q_r)\tau_r), \\ P &= 2K_2 + 3K_3 + K_f, \end{aligned} \quad (5)$$

where L is regulation game length and λ_r is an attempt rate per regulation game. Destination pace predicts that rate; it is not multiplied into exposure a second time. Rebounds, assists, steals, blocks, turnovers, and fouls follow the specified conditional count order.

PER is a derived forecast, $R = \mathcal{P}_v(\mathbf{B}, M, K)$, using recorded minutes, a versioned formula, and team/league normalizers K . Realized normalizers may define a retrospective target; preseason predictors use only available information, and unknown forecast normalizers are propagated through joint draws. No duplicate PER likelihood is added to the count model.

Use a prespecified minutes threshold $m_0 > 0$ for PER reporting and evaluation. Low-minute players remain in the likelihood for all primitive outcomes; the threshold does not become an input or remove their participation outcomes. PER is undefined at zero minutes, and summaries above m_0 describe an explicitly conditional population.

3 Model architecture

3.1 Opponent strength contextualizes each game

The same scoring line can provide different evidence against different opponents. We model team scoring efficiency from a competition environment, team offense, opponent defense, and venue; a separate component models pace. Offensive evidence uses opponent defense, and defensive evidence uses opponent offense. Appendix B specifies this proposed model, inspired by Pomeroy’s methodology note [10].

The model separates a competition’s scoring environment from team strength. Teams appearing in domestic and cross-border competitions connect league schedules. Disconnected schedules cannot identify their relative levels from game outcomes alone; bridging assumptions and uncertainty must remain explicit. NCAA and European ratings are not automatically on one absolute scale.

Data availability determines the tier: T2 uses dynamic team effects, T1 uses static effects, and T0 supplies league, tier, country, and age-tier context with unavailable-rating masks. T0 is the initial fallback and contains no opponent-adjusted efficiency estimate.

Freeze dated rating snapshots and tiers across auxiliary-data comparisons. T1/T2 estimates and fitting recipes use only pregame information; validate their efficiency and pace forecasts separately. Appendix B gives the model and provenance.

3.2 The reference game tower

The tower encodes each game using production, minutes, age, timing, and opponent context. Exposure and recency pooling summarize established performance and recent form; trends and evidence volume preserve information that averages discard (Figure 1).

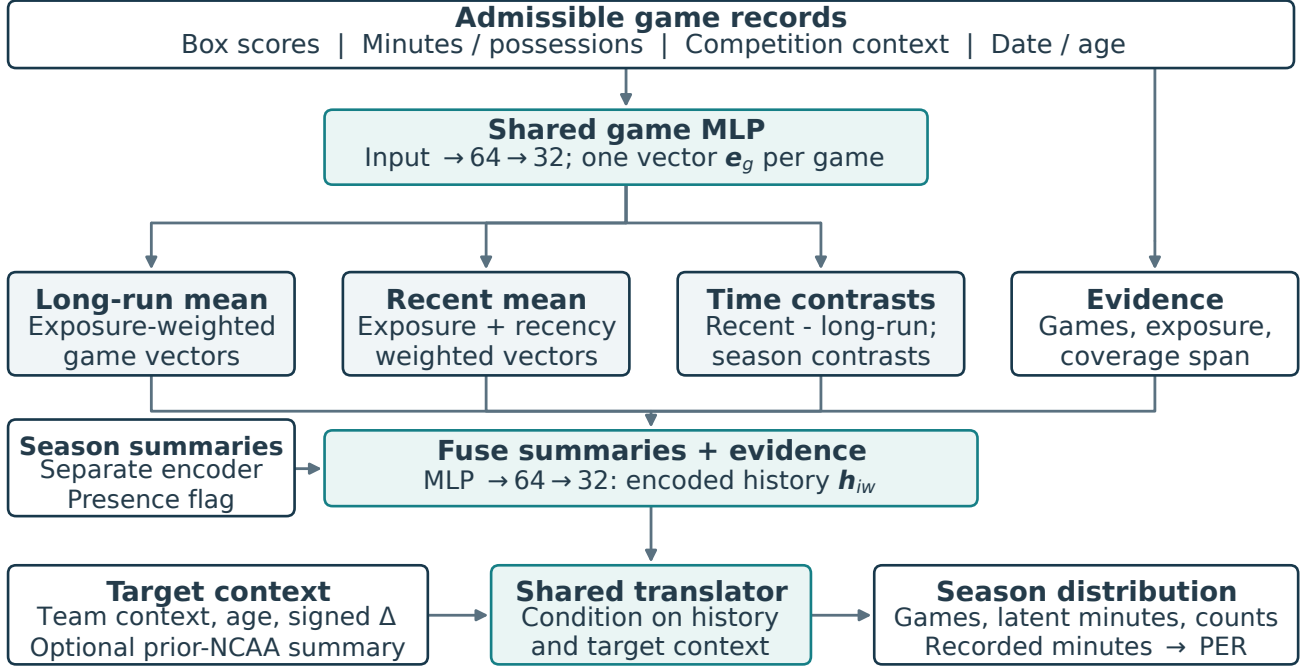


Figure 1: The reference tower and season forecast. Games share one encoder; weighted means, time contrasts, and evidence features summarize their history. Summary-only seasons have a separate encoder and presence flag; their seasons are disjoint from those in the game channel. Partial game coverage enters evidence features. Destination context joins downstream; prior NCAA games use a separate history channel. The reference contains no learned attention weights.

Let χ_{ig} contain transformed game features, field masks, category embeddings, and time features. The encoder is a small MLP with a 64-unit hidden layer and 32-dimensional output:

$$e_{ig} = \text{LN}\{W_2 \text{GELU}(W_1 \chi_{ig} + b_1) + b_2\} \in \mathbb{R}^{32}. \quad (6)$$

Retain target distance $\tau_g = s_g - T_t$, recency from the latest game in that channel, and the preceding-game gap. NCAA games use separate references; the translator's s_{iw} also admits summary endpoints. Professional-window recency therefore differs from distance to the NCAA target.

Let V_g identify valid games and $w_g = n_g + 1$, where n_g is observed player minutes. Then

$$h_i^L = \frac{\sum_g V_g w_g e_{ig}}{\sum_g V_g w_g}, \quad h_i^R = \frac{\sum_g V_g w_g 2^{-r_g/h_{1/2}} e_{ig}}{\sum_g V_g w_g 2^{-r_g/h_{1/2}}}. \quad (7)$$

Start with half-life $h_{1/2} = 0.5$ years, selected on forward validation. The added minute keeps documented DNPs represented. The time contrasts are $d_i^{RL} = h_i^R - h_i^L$ and masked within-/between-season contrasts projected to $d_i^{\text{season}} \in \mathbb{R}^{32}$.

Evidence features $\mathcal{E}_i$ include games, exposure, span, seasons, coverage completeness, and rating uncertainty. Fusion combines them with the summaries:

$$\mathbf{h}_{iw} = \text{LN}\{\text{MLP}_{64 \rightarrow 32}[h_i^L, h_i^R, d_i^{RL}, d_i^{\text{season}}, z_{iw}^{\text{sum}}, \mathcal{E}_i, m_i^{\text{hist}}]\}. \quad (8)$$

Here m_i^{hist} contains presence flags and z_{iw}^{sum} is the season-summary channel. Prior NCAA games produce a separate $\mathbf{h}_{it}^N$. Empty game channels bypass divisions; summary-only histories remain nonempty source histories. A wholly empty history returns a learned vector and zero evidence. Appendix C specifies the implementation.

3.3 Summary-only seasons

A complete season summary is encoded from totals, rates, exposure, competition, and period boundaries by a separate small network. Its vector joins fusion with a presence flag; it contributes no invented game dates or game-to-game contrasts. Admit it only when its whole period and release date satisfy the task’s cutoffs.

Each player-season enters at most one channel. Logs that reconcile with an available summary within a registered tolerance enter the game channel. A season without logs may enter the summary channel. Partial seasons retain their available games, record completeness in $\mathcal{E}_i$, and exclude the overlapping summary from its encoder. Reconciliation uses comparable periods and only information available at the task cutoff; unknown coverage is flagged. Appendix C gives the aggregation.

3.4 Translation and the shared learning path

The translator combines the history vectors with target age, NCAA experience, signed time, and the supplied team’s context:

$$u = f_{\Theta_{\text{trans}}}[\mathbf{h}_{iw}, \mathbf{h}_{it}^N, Z_{it}, C_{jt}^{(ct)}, v_{\text{time}}] \in \mathbb{R}^{32}. \quad (9)$$

It is a 64-hidden-unit MLP. The time features include source and target age, $|\Delta|$, and a small spline basis gated separately for negative and positive Δ . The reference uses one translator and one set of outcome heads; it has no separate directional adapter.

Reconstruction updates the shared encoders for observed channels, fusion, translator, and outcome heads. These parameters also support other players’ forward forecasts. Denote them by Θ ; inactive directional inputs can receive zero data gradient. Forward validation determines whether the shared updates help.

3.5 What is optional

Related internal NCAA results motivate the small reference model but do not validate this transfer task. Test the following alternatives separately; Appendix C.3 gives their equations.

Alternative	Question it tests
Scalar attention pooling	Does learning a weight per game improve fixed weighted means?
Temporal encoder and target-query reader	Do interactions between games or destination-dependent reading help?
Season hierarchy, adapters, roster reread	Does extra structure improve transfer enough to justify its estimation cost?
Game deletion, learned recency, richer context	Does regularization or additional information improve forward forecasts?

The supplied attention illustration appears as Figure 2: it depicts a separate 64-dimensional scalar-scoring variant, not the 32-dimensional reference or a destination-query reader.

4 Training and inference

4.1 Fit both directions; select on forward performance

For each direction, let $\bar{\ell}_d$ average the negative log season likelihood equally over players, then their target seasons and windows. This avoids allowing long careers and overlapping windows to dominate. The primary objective is

$$\mathcal{L}(\Theta) = \bar{\ell}_-(\Theta) + \lambda_+ \bar{\ell}_+(\Theta). \quad (10)$$

Choose λ_+ from a small prespecified grid such as $\{0, 0.1, 0.3, 1\}$ using earlier forward validation cohorts. Zero is a valid result. The raw number of reconstruction records does not set the weight. Optional pretraining and NCAA-only tasks are held constant across the main comparison; their complete objective is in Appendix D.

During fitting, use observed preceding counts, but integrate latent minutes rather than substituting recorded minutes. Training, calibration, and evaluation share one observed-data likelihood. Fit any calibration on earlier rolling out-of-fold forward forecasts, adjust valid conditional parameters, and regenerate draws. Independent corrections to points, attempts, and makes could violate scoring identities. Final test outcomes never enter fitting or selection.

4.2 Forecasts available to the user

For a supplied team, draw schedule length if needed, then games, starts, overtime, latent minutes, and counts. Draw recorded minutes through the same observation kernel; derive points and PER. Operational scoring marginalizes unknown schedule length, while realized- S scoring is a labeled scenario analysis. Report means, intervals, nonparticipation probability, and

$$\begin{aligned} \Pr(M \geq m_0 \mid X), \quad p(R \mid M \geq m_0, X), \\ \Pr(M \geq m_0, R > \tau \mid X) = \Pr(M \geq m_0 \mid X) \Pr(R > \tau \mid M \geq m_0, X). \end{aligned} \quad (11)$$

Estimate the joint event directly from draws. It is not the product of two unconditional marginal probabilities. Label m_0 according to its purpose: a PER reporting floor is not automatically a rotation threshold.

An updated later-season forecast adds NCAA games observed since the previous forecast. A longer-horizon forecast issued today retains today’s information cutoff and uses supplied or projected future team context. Separate season marginals do not imply a joint career distribution; that requires an explicit trajectory model.

4.3 The experiment that measures auxiliary-data value

Use rolling historical training cutoffs and evaluate on held-out first-year NCAA entrants with pre-NCAA international histories. Keep every evaluation player out of all fitting directions and pretraining. Report later-year outcomes separately. The main question is whether true NCAA correspondence helps beyond simply seeing extra game histories.

Model	Training data/procedure	Purpose
A	Forward supervision	Starting benchmark.
B	A plus history pretraining	Value of additional games without NCAA correspondence.
C	B plus true reconstruction pairs	Additional value of retrospective NCAA supervision.
D	C plus forward fine-tuning	Optional deployment refinement; give B the same opportunity.
C^{shuf}	B plus shuffled reconstruction pairs	Diagnostic for player correspondence.

B and C start from the same pretrained checkpoint and see the same additional international histories. Match architecture, forward/pretraining updates, preprocessing, rating inputs, calibration, and optimization/selection budgets. Evaluate shuffled supervision within prespecified relevance strata; beating a harmful shuffled model alone does not establish improvement over B.

For the same n forward test players, the primary contrast is

$$\hat{G}_+ = \frac{1}{n} \sum_{i=1}^n [\ell(\mathbf{Y}_i, \hat{p}_{B,i}) - \ell(\mathbf{Y}_i, \hat{p}_{C,i})]. \quad (12)$$

A positive value favors reconstruction supervision. Report nats per player on the same outcome definitions and units. Calibration does not make scores from different targets or populations comparable.

Use paired uncertainty estimates that preserve repeated-player and team-season dependence, and report individual cohort results when cohorts are few. Training-seed variability is a separate source of uncertainty: refit both arms and summarize paired differences. A three-seed range is an initial check, not a confidence interval. Assess log score, marginal error/CRPS, nonparticipation Brier score, interval coverage, and sharpness [15].

Repeat B versus C using only reconstruction players without pre-NCAA histories. Then test tower alternatives and ratings in separate controlled comparisons. This separates the contribution of auxiliary labels from architectural improvements. Registered decision rules, learning curves, and reproducibility checks are in Appendix D.

4.4 External comparators

Include hierarchical distributional regression, boosted-tree models with comparable opponent-adjusted summaries, and the latent-trajectory alternative. Joint-NLL comparisons require compatible normalized outcome distributions; point predictors are evaluated with point-error metrics.

A Glazer-inspired matched translation-factor comparator [6] is useful where comparable league movers and stayers exist. Compare rate adjustment at matched exposure, assess balance/overlap, and state the identifying assumptions. Forecasting season totals additionally requires opportunity and development models. All primary comparators obey the same historical availability rules.

5 Properties and limitations

Coherent outcomes. Normalized conditionals and recording kernels yield a normalized observed-outcome distribution. The model enforces $0 \leq J \leq G \leq S$, bounded latent minutes conditional on overtime, the declared recorded-minute support, nonnegative production, and makes no greater than attempts. Points are derived exactly. It does not enforce every possession constraint or a whole-roster minutes budget.

The right evaluation target. Expected forward negative log score is minimized by the true forward distribution [15]. Auxiliary fitting may reduce estimation error in shared relationships, but a mixed objective need not recover that distribution under misspecification. Appendix E gives the log-score identity and an idealized Fisher-information argument; neither guarantees a gain for this fitted neural model.

Forward labels remain necessary. Positive-time data alone do not identify the negative-time mapping. Professional successors and incoming NCAA recruits differ in age, selection, role, and competition. Shared representations or overlap weighting do not by themselves solve those differences [16]. Keeping $\lambda_+ = 0$ is preferable to harmful sharing.

Forecast uncertainty has several sources. The outcome likelihood captures season variability. Rating trajectories, schedules, and future team context contribute uncertainty only if propagated; parameter uncertainty needs separate assessment. Validate coverage by horizon and history volume. Destination scenarios remain conditional predictions, not causal estimates of switching teams.

Decision criterion. Adopt reconstruction supervision when it improves prospective entrant forecasts under a fixed, reproducible comparison and acceptable calibration. The paper’s contribution is this testable use of retrospective data, rather than a claim that added architecture or bidirectional training must help.

A Data and likelihood specification

This appendix makes the main model implementable without interrupting its narrative. Section 2 defines the forecast population and information cutoffs; the definitions below use those same conventions.

A.1 Admissible windows and task membership

Write $\mathcal{G}_i^I$ for games in the covered international competitions and $\mathcal{G}_i^N$ for NCAA games. A window contains a possibly empty game set $\mathcal{G}_{iw} \subseteq \mathcal{G}_i^I$ and encoded summary set $\mathcal{S}_{iw}$, with at least one record overall. Summaries used only to assess retained-game coverage form a disjoint set $\mathcal{S}_{iw}^{\text{cov}}$. Its games are admissible when

$$\begin{aligned} d = - : \quad & s_g < T_t, \quad \text{rel}(x_{ig}) \leq c_t \quad \forall g \in \mathcal{G}_{iw}, \\ d = + : \quad & s_g > E_t, \quad \text{rel}(x_{ig}) \leq c^{\text{tr}} \quad \forall g \in \mathcal{G}_{iw}. \end{aligned} \quad (13)$$

For a summary v in either summary set, covering $[s_v^{\text{start}}, s_v^{\text{end}}]$, require $s_v^{\text{end}} < T_t$ and release by c_t in the forward task; require $s_v^{\text{start}} > E_t$ and release by c^{tr} in reconstruction. An overlapping period is excluded, never split or assigned a midpoint. Define

$$s_{iw} = \max(\{s_g : g \in \mathcal{G}_{iw}\} \cup \{s_v^{\text{end}} : v \in \mathcal{S}_{iw} \cup \mathcal{S}_{iw}^{\text{cov}}\}). \quad (14)$$

Release cannot precede the information in a record. All fitting records and preprocessing also obey the training cutoff. Let $\mathcal{W}_{itd}$ contain admissible nonempty windows and $\mathcal{U}$ the verified roster-season outcomes released by c^{tr} . Then

$$\begin{aligned} \mathcal{D}_d &= \{(i, w, t) : (i, t) \in \mathcal{U}, w \in \mathcal{W}_{itd}\}, \\ \mathcal{I}_d &= \{i : \exists (w, t) (i, w, t) \in \mathcal{D}_d\}, \quad \mathcal{T}_d = \{t : \exists w (i, w, t) \in \mathcal{D}_d\}. \end{aligned} \quad (15)$$

$\mathcal{D}_N$ contains NCAA-only history-to-future-season units with no international window and nonempty available prior NCAA history; it is separate from $\mathcal{D}_-$. A wholly empty source has no signed source date: mask source-age/time features rather than assign a direction. A summary-only source remains nonempty under (14).

Reconstruction views obey a registered horizon and the training release cutoff. Mark partially observed game-log seasons and deduplicate on game and both summary-set IDs after clipping. Complete summaries still obey whole-period admission. Compute player/season/window loss weights after deduplication; the initial window choices appear in Appendix D.2.

The common pool $\mathcal{P}$ contains fold-admissible histories from eligible international-only and NCAA-linked players, excluding every evaluation player. Hide NCAA correspondence, prior NCAA inputs, destination context, target age/experience, Δ , and target-anchored game clocks. Use source-only reference dates. A next-game objective recomputes all prefix clocks, exposure and season summaries from information then available; future complete-season summaries cannot enter its prefix. Register one pretraining recipe and reuse the same records, initialization, objective, and checkpoint for B, C, and C^{shuf} .

Pre intl.	NCAA label	Post intl.	Task contribution
Yes	Yes	No	$\mathcal{D}_-$: direct forecast supervision.
No	Yes	Yes	$\mathcal{D}_+$: reconstruction without a pre-NCAA history.
Yes	Yes	Yes	Both tasks, sharing the observed NCAA outcome.
Yes	No	Yes	History/development pretraining, without an invented NCAA label.
No	Yes	No	$\mathcal{D}_N$ if a prior NCAA window exists.

A senior-season label does not become a first-year label when reconstructed. An observed first-year label can pair with a later professional window, with its actual larger positive Δ . Records of unresolved roster identity are held outside confirmed outcome units pending resolution; they are not automatically zero appearances.

A.2 Starts, overtime, and latent minutes

For $g > 0$, use the following conditional distributions:

$$\begin{aligned} J \mid g, u &\sim \text{BetaBin}(g, \alpha_J, \beta_J), & O \mid g, u &\sim \text{Poisson}\{g\lambda_O(u)\}, \\ U \mid g, o, j, u &\sim \pi_M \delta_1 + (1 - \pi_M) \text{Beta}(\alpha_M, \beta_M), \\ \overline{M}(g, o) &= Lg + L_{OT}o, & M^* &= \overline{M}(g, o)U. \end{aligned} \quad (16)$$

All head parameters may depend on the conditioning variables shown. Set $O = 0$ on the zero-game branch. Overtime is summed out when unobserved. Its Poisson distribution is untruncated; numerical summation and sampling must use that same model.

For any probability parameter use $\text{sigmoid}(h) = 1/(1 + e^{-h})$; for a positive parameter use $\text{softplus}(h) = \log(1 + e^h)$, with numerical safeguards. Beta-binomial shape parameters and negative-binomial dispersions remain positive. $\text{NegBin}(\mu, \kappa)$ has variance $\mu + \mu^2/\kappa$. The displayed factorization makes starts and overtime conditionally independent given games and u ; a different dependence would require a corresponding factor change.

A.3 Recorded minutes and the observed likelihood

Let ϱ identify provider, recorded precision, and aggregation convention. The observation kernel is a conditional probability on the declared discrete reporting support $\mathcal{M}_{\varrho, g, o}$:

$$K_{\varrho, g, o}(m \mid m^*) = \Pr(M = m \mid M^* = m^*, g, o, \varrho), \quad \sum_{m \in \mathcal{M}_{\varrho, g, o}} K_{\varrho, g, o}(m \mid m^*) = 1. \quad (17)$$

The model assumes this reporting mechanism is independent of production conditional on latent minutes, games, overtime, and provenance. For $g > 0$, the observed contribution is

$$q(m, \mathbf{b} \mid g, j, u, \varrho) = \sum_{o=0}^{\infty} p(o \mid g, u) \int p(\mathbf{b} \mid m^*, g, u) K_{\varrho, g, o}(m \mid m^*) dF_{M^*|g, j, o, u}(m^*). \quad (18)$$

Multiply by $p(g \mid u, S)p(j \mid g, u)$ for the complete observed-data probability. The integral contains both the continuous minute fraction and its atom. The atom contributes $\pi_M K_{\varrho, g, o}(m \mid \overline{M}) p(\mathbf{b} \mid \overline{M}, g, u)$; it does not replace continuous mass compatible with the same recorded value. The $G = 0$ branch remains deterministic.

Recording conventions. The kernel follows a documented provider rule or a validated working approximation. A once-quantized total uses the indicator of its reporting cell. Sums of quantized game minutes additionally require a conditional model for per-game allocations or errors. For example, a centered Irwin–Hall error approximation assumes independent uniform per-game errors and has variance $g/12$; integer storage alone does not establish these assumptions. Each kernel normalizes on its declared reporting support and handles endpoints. Provenance rules, approximations, and numerical tolerances are registered before use, with sensitivity analysis for plausible alternatives.

Training, calibration, and evaluation use the same observed-data function, evaluating production throughout the integral and retaining endpoint atoms exactly. Numerical integration is checked against an independent high-accuracy reference, including concentrated beta shapes and rare outcomes. The implementation plan specifies accuracy budgets and convergence checks.

Overtime stopping rule. Let q_N be the exact partial sum of (18) through $o = N$, and $T_N = \Pr(O > N \mid g, u)$. The recording kernel is a discrete probability. Production $\mathbf{B}$ is also discrete, so $p(\mathbf{b} \mid m^*, g, u) \leq 1$. Both factors in the integrand are therefore at most one, giving for $q_N > 0$,

$$0 \leq q - q_N \leq T_N, \quad 0 \leq \log q - \log q_N \leq \log(1 + T_N/q_N). \quad (19)$$

An absolute tail threshold alone does not control log-score error for rare observations. Equation (19) bounds omission of the untruncated Poisson tail for exact q_N ; numerical quadrature error is assessed separately. Forecast simulations use the same untruncated overtime distribution and recording kernel as the likelihood.

A.4 Primitive production and PER conventions

The autoregressive vector is

$$\mathbf{B} = (A_2, K_2, A_3, K_3, A_f, K_f, \text{ORB}, \text{DRB}, \text{AST}, \text{STL}, \text{BLK}, \text{TOV}, \text{PF}). \quad (20)$$

Each component conditions on M^* , u , and earlier components; include G where its support is needed. Attempts and non-shooting counts follow the negative-binomial form in (5). Makes use their corresponding attempts as a beta-binomial denominator. Fouls are truncated to $0:fG$, with rule-based per-game limit f . Total rebounds, total field goals, and points are deterministic functions. All primitive counts are zero at $G = 0$; numerical positivity safeguards apply to rates, not an artificial positive mean on that branch.

The PER formula version and normalizer definitions must be fixed across models. If normalizers are unavailable, an explicitly separate approximation can replace the deterministic PER map with $R \mid \mathbf{B}, M, u \sim t_{\nu_R}(\mu_R, s_R)$, $\nu_R > 2$. Here s_R is scale, with SD $s_R \sqrt{\nu_R/(\nu_R - 2)}$. Such an approximation changes the model and score definition; it is not an extra independent score for deterministically duplicated data. The reference model derives PER from box scores and normalizer draws.

B Opponent-adjusted team ratings

Player translation needs an upstream description of the competition faced. We propose a *KenPom-style*, possession-normalized European rating model inspired by adjusted offense/defense and additive opponent corrections [10]. This is our specification, not a reproduction of KenPom’s proprietary implementation. Country, league, and raw scoring average are not interchangeable measures of competitive strength.

For game g at date s , let A, B be teams, Q_g possessions, $\text{env}(g)$ the competition/rules environment, and h_g a signed venue indicator. Set $y_{Ag} = 100P_{Ag}/Q_g$, and model the two scoring efficiencies jointly:

$$\begin{aligned} \begin{pmatrix} y_{Ag} \\ y_{Bg} \end{pmatrix} &\sim t_{\nu} \left(\begin{pmatrix} \mu_{\text{env}(g),s} + o_{A,s} - d_{B,s} + \beta h_g \\ \mu_{\text{env}(g),s} + o_{B,s} - d_{A,s} - \beta h_g \end{pmatrix}, \frac{Q_{\text{ref}}}{Q_g} \Sigma_{\text{env}(g)} \right), \\ \log \tilde{Q}_g &\sim N \left(\kappa_{\text{env}(g),s} + \frac{1}{2}(v_{A,s} + v_{B,s}), \sigma_Q^2 \right). \end{aligned} \quad (21)$$

Here t_{ν} uses a scale matrix; Q_{ref} is a fixed reference exposure. $\tilde{Q}_g$ is possessions normalized to a common regulation duration, including overtime adjustment. Positive o denotes better offense; positive d denotes better defense (fewer points allowed); v denotes pace preference. Thus reference-environment ratings are $\text{AdjO}_A = \mu_{\text{ref}} + o_A$, $\text{AdjD}_A = \mu_{\text{ref}} - d_A$. Distinguish the intercept μ_c , a scoring environment, from team strength.

Use hierarchical initial states and time evolution, for example

$$\xi_{A,s} = \xi_{A,s'} + \epsilon_{A,s}, \quad \epsilon_{A,s} \sim N(0, (s - s')\Omega), \quad \xi = (o, d, v). \quad (22)$$

Season breaks may receive larger transition variance or roster-change covariates. Hierarchical priors shrink sparse teams toward league/age-tier means. Center team effects and fix reference intercepts; constrain any league-plus-team decomposition to avoid redundant offsets. Estimate from team games independently of the player-outcome decoder, then freeze ratings for the main transfer experiment.

Cross-league alignment. Teams appearing in domestic and cross-border competition connect schedules; use a common team-season identity, while accounting for material roster differences. These bridges help

estimate relative levels. Disconnected components do not identify their offsets from game results alone. Common-player transitions may provide additional bridges only under an explicit development/selection model. NCAA and European ratings are not automatically on one absolute scale: retain domain indicators, regularized bridge parameters, and uncertainty; NCAA transition labels calibrate the player translation.

Data provenance and uncertainty. Record provider/retrieval time, competition/age tier, IDs, dates, venue, scores, minutes, game length, and possession definition. Prefer play-by-play possessions; version approximations and flag measurement error. Deduplicate cross-competition logs and validate youth/reserve/senior labels.

Use *pregame* rating means, uncertainties, and coverage masks. A fold snapshot records the training cutoff, source-data dates, rating-fit recipe, and version. T1 fits static team effects on information available before each source game; T2 filters dynamic states using that same historical restriction. Parameters and tuning used to produce a claimed pregame value must also respect its information cutoff. A static fit on all games before c^{tr} is not automatically a pregame fit for earlier source games. Lagged NCAA destination ratings require their own verified availability by c_t .

Correct uncertainty propagation requires coherent rating-trajectory posterior draws; standard-error features alone are insufficient. Freeze tiers and snapshots across auxiliary-data ablations, and validate T1/T2 efficiency/pace forecasts separately. T0 uses catalog descriptors and masks without claiming numerical team strength or efficiency/pace validation.

C Game-tower implementation and alternatives

C.1 Features, clocks, and history summaries

Let a batch contain B_{batch} examples, at most T games each, and F continuous fields. Valid-game mask V and field masks $\mathbf{m}$ are distinct. In training-fold transformations, use $\tilde{x}_f = \mathbf{m}_f\{T_f(x_f) - \mu_f\}/(\sigma_f + \epsilon)$ and append $\mathbf{m}_f$. Use log transforms for skewed nonnegative counts, not signed ratings. Include attempts alongside shooting proportions and preserve exposure denominators. Categorical descriptors use 8-dimensional embeddings with an unknown-category entry. No player-identity embedding memorizes labels.

The translator uses source age $a_{iw} = a_i(s_{iw})$, target age a_{it} , and

$$v_{\text{time}} = [a_{iw}, a_{it}, |\Delta|, \{\mathbf{1}(\Delta < 0)B_k(|\Delta|), \mathbf{1}(\Delta > 0)B_k(|\Delta|)\}_{k=1}^{K_b}], \quad (23)$$

with a small spline basis. Mask undefined source-age/time entries for NCAA-only or empty-history inputs. The two ages satisfy (1).

The time features are

$$[\tau_g, \text{sgn}(\tau_g) \log(1 + |\tau_g|), \log(1 + r_g), \log(1 + \delta_g), \text{season-boundary mask}], \quad (24)$$

where $\tau_g = s_g - T_t$, r_g is elapsed time to the latest game in the same game channel, and δ_g is its preceding-game gap. For international games, set $s_{iw}^G = \max_{g \in \mathcal{G}_{iw}} s_g$ when that set is nonempty and $r_g = s_{iw}^G - s_g$. This game anchor can differ from the summary-inclusive s_{iw} in (14). The NCAA channel has its own anchor and gaps. Compute them before row permutation, include first-game flags, and bypass empty-channel calculations. Pooled predictions agree up to floating-point effects when complete records are reordered.

Within the latest season, subtract the mean embedding of the preceding five games from that of the final five, requiring at least ten games. The between-season contrast subtracts the preceding season's weighted mean from the latest season's, requiring both. Project their concatenation, with presence masks, to $d_i^{\text{season}} \in \mathbb{R}^{32}$. Absent contrasts use zero vectors plus flags.

Evidence features include valid games, observed minutes, coverage completeness, window span, gap to the target, seasons, field coverage, and rating uncertainty. Where an eligible summary permits reconciliation over the same period, record observed and reported game counts, their coverage ratio, and reconciliation

status. Unavailable denominators receive an unknown-coverage flag. Partial logs remain game inputs; their summary is used only for eligible coverage metadata and excluded from the summary encoder. Channel assignment precedes game-deletion augmentation, which cannot move a season into the summary channel. If minutes are unavailable for an input game, a fold-specific typical exposure can be used only for pooling weights, with its mask; compare equal weights in sensitivity analysis. The raw field retains its missing-value mask. Keep game-log and summary evidence volumes distinct; wholly empty histories bypass divisions and return the learned empty vector.

For each admitted summary v , let ζ_{iv} contain transformed totals/rates, exposure, masks, competition descriptors, and period start/end features. A separate 64-hidden-unit MLP produces a 32-dimensional vector. With reported season minutes m_{iv}^{sum} and weight $b_{iv} = 1 + m_{iv}^{\text{sum}}$, the initial aggregation is

$$z_{iw}^{\text{sum}} = \frac{\sum_{v \in \mathcal{S}_{iw}} b_{iv} \text{MLP}_{64 \rightarrow 32}^{\text{sum}}(\zeta_{iv})}{\sum_{v \in \mathcal{S}_{iw}} b_{iv}}. \quad (25)$$

Use a zero vector and absence flag for an empty summary set. Require verified exposure for this weighted version; an explicitly flagged alternative can use equal weights. Summary-only seasons contribute no invented per-game gaps or game contrasts, and never duplicate a season already represented by game logs in the reference channel. Record its capacity in every comparison; pretraining arms use the same enabled channels.

C.2 The supplied attention illustration

Figure 2 shows a scalar attention-pooling alternative. For a learned score $a_{ig} = s_{\vartheta}([e_{ig}; \tilde{x}_{ig}])$, compute

$$\alpha_{ig} = \frac{V_g \exp(a_{ig})}{\sum_h V_h \exp(a_{ih})}, \quad h_i^A = \sum_g \alpha_{ig} e_{ig}. \quad (26)$$

Implement this with stable masked softmax and bypass it for empty histories. This alternative learns one scalar weight per game, without pairwise self-attention or a destination query. To compare it with the reference at fixed width, keep $D = 32$, append h_i^A to fusion, and register the extra capacity. Replacing the two weighted means is a separate ablation. Dates may affect scores through features; learned weights need not decrease monotonically with age.

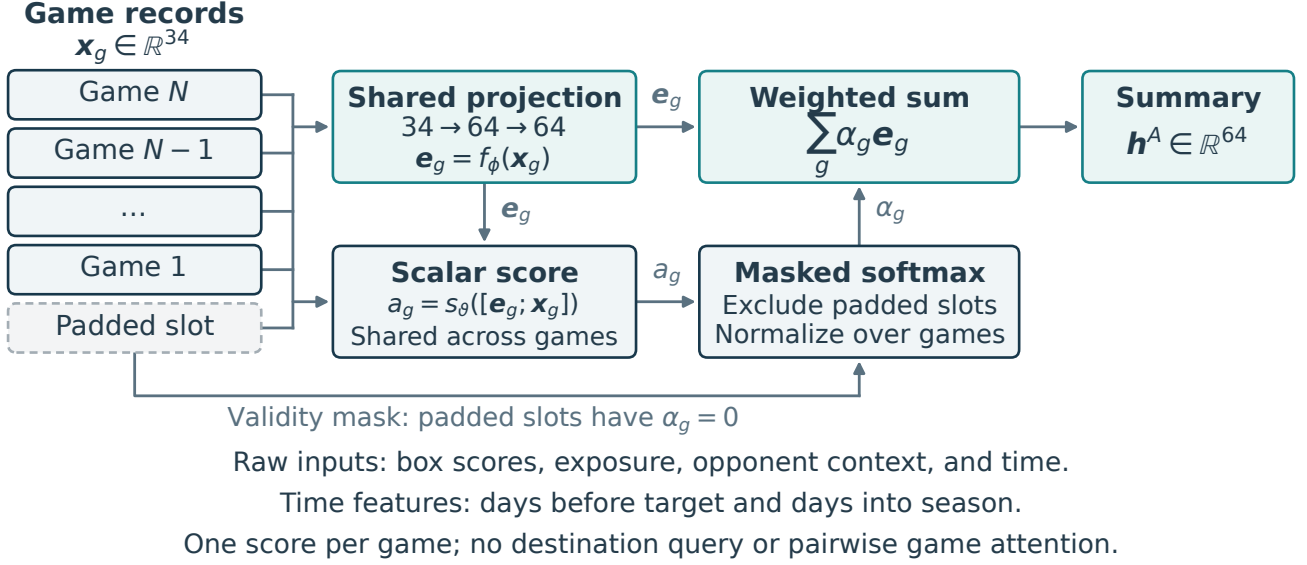


Figure 2: Scalar attention pooling, redrawn from the supplied illustration for print. Its configuration uses 34 input fields and 64-dimensional embeddings. Embeddings supply the values in the weighted sum; the scoring branch supplies one weight per real game and zero weight on padding. This is an optional variant, not the reference model in Figure 1, and it contains no destination-conditioned query.

C.3 Registered architectural challengers

In this appendix $D = D_u = 32$, $d_h = D/4$, and e_{ig} is abbreviated to e_g where unambiguous. Each component below is fully specified so that it can be tested, and none is part of the reference model. A challenger is adopted only if it improves forward scores on held-out entrants beyond paired evaluation uncertainty under Section 4; a null result retains the simpler model. Test each independent change against the reference. Components that require another challenger are subsequent interaction tests, with the dependency retained in both comparison arms. Test interactions with reconstruction supervision separately (Section 4.3).

(a) Temporal self-attention over games. Add a contextualized branch using two pre-normalized temporal self-attention blocks, each with four heads of width $d_h = D/4$ and a $D \rightarrow 2D \rightarrow D$ feed-forward sublayer [11]. For one head and valid query game g ,

$$\begin{aligned}
A_{gh}^{(k)} &= \frac{(W_Q^{(k)} \bar{e}_g)^\top (W_K^{(k)} \bar{e}_h)}{\sqrt{d_h}} + b^{(k)}(\rho_{gh}) + \mathcal{M}_{gh}, \\
\rho_{gh} &= [\text{sgn}(s_g - s_h), \log(1 + |s_g - s_h|), \mathbf{1}(\text{season}_g = \text{season}_h)], \\
o_g^{(k)} &= \sum_h \text{softmax}_h(A_{gh}^{(k)}) W_V^{(k)} \bar{e}_h, \quad \bar{e} = \text{LN}(e), \\
y &= e + \text{Drop}\{W_O[o^{(1)}; \dots; o^{(4)}]\}, \quad z = y + \text{Drop}\{\text{FFN}(\text{LN}(y))\}.
\end{aligned} \tag{27}$$

Each $b^{(k)}$ is a small learned time-bias MLP. This continuous-time adaptation is motivated by relative-position attention [13], not identical to that paper’s discrete-position construction. Use dropout 0.1 on residual branches and the feed-forward hidden layer; pass the first block’s output to the second. Set $\mathcal{M}_{gh} = -\infty$ for invalid keys and zero otherwise; padded queries are skipped and zeroed after each block. All observed games may interact because they are available for that example. Next-game pretraining additionally masks later games and recomputes prefix features, avoiding future exposure or season-summary leakage. A one-layer chronological GRU with D hidden units [12] is the recurrent variant of this challenger.

The explicit summaries (7) continue to use the *pre-temporal* embeddings e , so their interpretation is not blurred by game-to-game mixing. When tested alone, pool z with the exposure weights of (7) and append this summary to fusion. Challenger (b) is a separately registered readout replacement; its combination with (a) is a subsequent interaction test. Cost is $O(T^2D)$ against $O(TD)$ for the reference.

(b) Destination-conditioned history reader. Form a query from lagged destination ratings and masks, target age, NCAA experience, horizon, and player attributes:

$$q_i = \text{MLP}_{2D \rightarrow D}[C_{jt}^{(ct)}, Z_{it}, \Delta, |\Delta|, \mathbf{1}(\Delta > 0)] \in \mathbb{R}^D. \quad (28)$$

The subscript gives hidden/output widths; input width is determined by the fields. With four query-attention heads over game embeddings z_{ig} (or e_{ig} without challenger (a)),

$$\begin{aligned} a_{ig}^{(k)} &= \frac{(U_Q^{(k)} q_i)^\top U_K^{(k)} z_{ig}}{\sqrt{d_h}} + \tilde{b}_k(\tau_g, r_g) + \mathcal{M}_g, & \alpha_{ig}^{(k)} &= \text{softmax}_g a_{ig}^{(k)}, \\ h_i^{A,k} &= \sum_g \alpha_{ig}^{(k)} U_V^{(k)} z_{ig}, & h_i^A &= \text{LN}\{U_O[h_i^{A,1}; \dots; h_i^{A,4}]\}. \end{aligned} \quad (29)$$

Each head returns d_h coordinates, giving $h_i^A \in \mathbb{R}^D$, which is appended to the fusion input in (8). $\mathcal{M}_g$ excludes padding and inadmissible games. Opponent ratings influence keys and values through z_{ig} , so equal points against different defenses need not mean equal performance. Destination pace/style can change which evidence matters. We do not force recent games or strong opponents always to receive higher weights, nor interpret weights causally. Set Transformer [14] provides precedent for attention pooling; our query is specifically supplied by the forecast scenario. Add $N_{\text{att}}^{(k)} = 1/\sum_g (\alpha_{ig}^{(k)})^2$ per head to $\mathcal{E}_i$ as an attention-concentration diagnostic, *not* an independent sample size. For one game the attention weights are one.

(c) Season hierarchy. A single softmax can dilute a short recent season among many old games. Restrict the reader in (29) to each nonempty season s , obtaining h_{is}^A . Let η_{is} contain season age, signed target distance, minutes, games, average opponent ratings, and masks. Pool across seasons:

$$\beta_{is} = \text{softmax}_s \left\{ w^\top \tanh(W_h h_{is}^A + W_q q_i + W_\eta \eta_{is}) \right\}, \quad h_i^S = \sum_s \beta_{is} h_{is}^A \in \mathbb{R}^D, \quad (30)$$

appended to the fusion input. This learns season weights from both content and context. Use documented source season definitions, retain dates across differing competition calendars, and exclude empty seasons from season attention.

(d) Directional adapters. Add rank-16 residual adapters after the shared translator:

$$u_d = u + A_d \text{GELU}(B_d u + b_d), \quad B_d \in \mathbb{R}^{16 \times D_u}, \quad A_d \in \mathbb{R}^{D_u \times 16}, \quad d \in \{-, +\}, \quad (31)$$

with adapter parameters $\psi_d = (A_d, B_d, b_d)$ and the outcome heads consume u_d . Add $\rho(\|\psi_-\|_2^2 + \|\psi_+\|_2^2)$ to (10), where ψ_d collects the adapter parameters. The forward adapter is trained only by the scarce forward sample, which is precisely the setting in which direction-specific modules are prone to overfit; it must demonstrate forward value against the shared network with directional bases alone.

(e) Roster reread. If a preseason destination roster is supplied, a Portal Machine-style reread uses self-excluded teammate context $c_{-i,j} = \text{Pool}\{\mathbf{h}_l : l \in \mathcal{R}_{jt}, l \neq i\}$, with $\mathcal{R}_{jt}$ released by c_t and $\mathbf{h}_l$ each teammate's fused history, and $q'_i = q_i + \text{MLP}[q_i, c_{-i,j}]$. Recompute the reader with q'_i and add a gated residual to $\mathbf{h}_{iw}$. Teammate summaries come from an initial pass, excluding the subject; avoid circular simultaneous updates. Every roster-derived quantity must be recomputed from the supplied scenario roster, never from a realized roster. Disable this extension when only ORtg/DRtg/pace are available. It does not itself impose a whole-roster minutes budget.

(f) Game deletion. During training, retain each valid game independently with probability $1 - p_G$, recomputing the pooling denominators in (7), the contrasts, and the evidence summaries $\mathcal{E}_i$ from the retained games only; evaluation uses all games. Test $p_G \in \{0.1, 0.2, 0.3\}$ as a regularizer candidate. Shuffling is not an augmentation for the reference tower, which is order invariant by construction.

(g) Candidate inputs. A learned half-life $h_{1/2} = h_{\min} + \text{softplus}(\eta)$, with fixed $h_{\min} > 0$, in (7); the player’s role within the source team (shares of team points, attempts, and usage; minutes rank on the team; number of contributors); and a hierarchically shrunk destination-conference embedding in $C_{jt}^{(c_t)}$. Reported gains for these inputs in related NCAA models do not establish gains here.

C.4 Latent trajectories and paired-history teaching

A complementary comparator separates development from league-specific observation, following nonlinear state-space modeling [8]. For chronological player-season records $k = 1, \dots, K_i$, let ζ_{ik} be a latent player state, a_{ik} age, h_{ik} elapsed time, L_{ik} league, and D_{ik} opportunity/production observations. A candidate joint model is

$$p(\zeta_{i1}) \prod_{k=2}^{K_i} p_\phi(\zeta_{ik} \mid \zeta_{i,k-1}, a_{ik}, h_{ik}) \prod_{k=1}^{K_i} p_\psi(D_{ik} \mid \zeta_{ik}, L_{ik}, C_{ik}). \quad (32)$$

Use league-specific structured emission heads on the available chronological records. A variational smoother may use the available whole training trajectory; deployment infers states from the observed prefix and propagates them forward. Thus an NCAA/post-international trajectory contributes without pre-NCAA observations. Count each observed record once in this joint model. Chronological transitions use positive elapsed time; backward smoothing is posterior inference, not a negative-time dynamical inverse. Latent states are not identified intrinsic ability without additional restrictions.

The game tower may parameterize the inference network, but (32) is an alternative to the primary conditional estimator, not an assumed equivalence. For complete paired histories, future-aware teacher representations or soft predictions may additionally train a past-only student [2, 3, 7]. Evaluate this extension separately; forcing equal predictive distributions across different information sets can miscalibrate forecasts. Missing student histories prevent ordinary paired distillation from using the dominant reconstruction-only group directly.

D Training details and extended comparisons

D.1 Player-balanced loss and optional tasks

For nonempty direction- d sample, define

$$\bar{\ell}_d = \frac{1}{|\mathcal{I}_d|} \sum_{i \in \mathcal{I}_d} \frac{1}{|\mathcal{T}_{id}|} \sum_{t \in \mathcal{T}_{id}} \frac{1}{|\mathcal{W}_{itd}|} \sum_{w \in \mathcal{W}_{itd}} [-\log p_{d,\Theta}(\mathbf{Y}_{it} \mid X_{iwt}, S_{jt})]. \quad (33)$$

This is a conditional/composite objective over dependent windows, not an independent joint career likelihood. The extended objective adds $\lambda_N \bar{\ell}_{N \rightarrow N} + \lambda_U \mathcal{L}_{\text{pretrain}}$ to (10), and an adapter penalty only when that challenger is active. Optional task definitions and their weights are fixed across B and C. Omit an absent task rather than divide by an empty sample.

D.2 Calibration and reproducibility

Calibrate raw out-of-fold conditional predictions once. Hurdle probabilities and conditional beta-binomial means can be adjusted on logit scales; rate and concentration/dispersion parameters on log scales. The minute-fraction model needs its own support-preserving parameter calibration. Regenerate draws and check derived outcomes. Recipe, available inputs, outcome units, and evaluation population remain identical across compared models.

Record data snapshots, cutoffs, preprocessing, rating versions, splits, seeds, hyperparameters, and exact checkpoints. Fixed-checkpoint predictions establish reproducibility; retraining variability is reported separately. The auxiliary data-loss gradient on a forward-only input weight is zero, although optimizer state and other loss terms can still change that parameter.

The reference starts with $D = D_u = 32$, hidden widths 64, residual/hidden dropout 0.1 where applicable, and $h_{1/2} = 0.5$ years. Use at most four prespecified spline basis functions per direction, with locations fixed from training data. Bucket by history length. Game aggregation costs $O(TD)$; summary aggregation is linear in admitted summaries; pairwise game attention costs $O(T^2D)$. Predeclare lookback caps and decision criteria. Under game deletion, rebuild game summaries, coverage, and source anchors consistently while keeping the forecast cutoff fixed. Existing summary-only seasons remain separate; deletion does not synthesize a full-season summary from removed games.

Implementation defaults. Season assignment uses, in order, a verified provider competition-season identifier, the competition’s period boundaries, and the 1 August fallback; records whose assigned season differs from their original label are listed for audit rather than silently relabelled. Forecasts use October 1 of the opening year, with training the previous day: season 2026 has $c_t = 2025-10-01$ and $c^{\text{tr}} = 2025-09-30$. When publication dates are unavailable, release is assumed to be the game date or completed summary endpoint plus one day. This assumption does not establish database ingestion time. Reconstruction views use the first $k \in \{1, 2, 3\}$ observed post-NCAA international seasons, capped at four years after E_t and clipped to the training release cutoff; complete summaries retain whole-period admission.

The protocol develops models in rolling folds through 2023, selects among at most three preregistered finalists on 2024–2025, and evaluates 2026 once after specifications and comparisons are frozen. B and C^{shuf} remain fixed comparison and diagnostic controls, respectively, without counting as selectable finalists. Calibration uses earlier out-of-fold predictions, and small subgroup results are descriptive. Empirical evaluation requires verified dates, complete outcomes, a registered recording kernel, and historical availability.

D.3 Learning curves and forward-equivalent data

Vary unique forward and reconstruction players, with repeated subsamples. If $R_B(n)$ is the baseline risk curve and $R_C(n_-, n_+)$ the augmented risk, define

$$n_{\text{eq}} = \inf\{n : R_B(n) \leq R_C(n_-, n_+)\}, \quad V_+ = n_{\text{eq}} - n_-, \quad m_+ = n_{\text{eq}}/n_-. \quad (34)$$

Interpolate only over the observed learning curve and report uncertainty. These are empirical comparisons, not universal effective sample sizes. If inversion is unstable, report the direct paired score difference. With very few cohorts, show cohort-level results and avoid interpreting resampling as new independent information.

D.4 Optional adaptive auxiliary weighting

This is an extension, not the primary procedure: learning task weights from a very small forward validation sample can itself overfit, so the prespecified grid in Section 4 is the default. Auto-Lambda [9] motivates learning auxiliary influence using a forward validation objective. With other loss terms suppressed for clarity, a one-step approximation is

$$\begin{aligned} \Theta'(\lambda) &= \Theta - \alpha \nabla_{\Theta} [\bar{\ell}_{-}^{\text{train}}(\Theta) + \lambda \bar{\ell}_{+}^{\text{train}}(\Theta)], \\ \lambda' &= \Pi_{[0, \lambda_{\text{max}}]} \left\{ \lambda - \eta \partial_{\lambda} \bar{\ell}_{-}^{\text{meta}}(\Theta'(\lambda)) \right\}. \end{aligned} \quad (35)$$

Here α, η are step sizes and Π projects onto a fixed bounded interval. The meta sample contains forward players available within the training-era cutoff and excluded from inner fitting. Reserve a separate forward validation sample for stopping and model selection; the test set never updates weights. Compare this extension with a tuned fixed λ_+ and zero. Start with one auxiliary weight: league/age-specific weights

require more validation data. Better auxiliary fit is not the optimization target, and adaptive weighting does not guarantee protection against negative transfer.

E Formal properties

E.1 Normalization and the forward log score

First sum over recorded minutes using (17), which equals one for each latent minute value. Then sequentially integrate or sum the normalized factors of (3). The zero-game branch is a point mass; bounded latent fractions and conditional makes enforce their supports. Integrating latent minutes and summing overtime preserve normalization. Deterministic scoring maps preserve the stated identities. Numerical approximation requires the separate accuracy controls in Appendix A.3.

For the true forward conditional p_- and a candidate q_- , with a common dominating measure and finite expectations,

$$R_-(q_-) - R_-(p_-) = \mathbb{E}_{X \sim P_-} [\text{KL}\{p_-(\cdot | X) \| q_-(\cdot | X)\}] \geq 0. \quad (36)$$

This follows by subtracting conditional expected log scores. It establishes the evaluation target, not consistency of a misspecified mixed-task estimator.

E.2 Idealized information gain

For this derivation only, use an idealized, correctly specified regular finite-dimensional model and the local parameter partition below; it is not a claim that every neural parameter is shared equally. Let θ be shared parameters, ψ_- forward-specific parameters, and ψ_+ reconstruction-specific parameters. Suppose independent forward and auxiliary players are sampled, and the forward Fisher information I_- for (θ, ψ_-) is positive definite. After profiling out ψ_+ , auxiliary data provide efficient shared information

$$I_+^{\text{eff}} = I_{\theta\theta}^+ - I_{\theta\psi_+}^+ (I_{\psi_+\psi_+}^+)^{-1} I_{\psi_+\theta}^+ \succeq 0. \quad (37)$$

For efficient joint likelihood estimation, the first-order covariance satisfies

$$V_{-+} \approx \left[n_- I_- + n_+ \begin{pmatrix} I_+^{\text{eff}} & 0 \\ 0 & 0 \end{pmatrix} \right]^{-1} \preceq (n_- I_-)^{-1} \approx V_-. \quad (38)$$

Matrix inversion reverses this information order; the delta method transfers it to local prediction-estimation variance. Strict gain requires prediction-relevant auxiliary information. This does not reduce irreducible variability or guarantee gains under weighted neural fitting, dependent views, or misspecification.

References

- [1] R. Caruana (1997). Multitask learning. *Machine Learning*, 28, 41–75. Author manuscript.
- [2] D. Serdyuk et al. (2018). Twin Networks: Matching the Future for Sequence Generation. *ICLR*. Paper.
- [3] Y. Feng, S. Gu, D. Guo, Z. Yang, and C. Shao (2021). Guiding Teacher Forcing with Seer Forcing for Neural Machine Translation. *ACL-IJCNLP*. Paper.
- [4] A. Goyal, A. Sordoni, M.-A. Côté, N. R. Ke, and Y. Bengio (2017). Z-Forcing: Training Stochastic Recurrent Networks. *NeurIPS*, 30. Publisher.
- [5] R. Sennrich, B. Haddow, and A. Birch (2016). Improving Neural Machine Translation Models with Monolingual Data. *ACL*, 86–96. Publisher.
- [6] A. K. Glazer (2026). Nuthin’ But A G League: Estimating league translation factors. *Journal of Sports Analytics*, online March 2. Publisher.
- [7] D. Lopez-Paz, L. Bottou, B. Schölkopf, and V. Vapnik (2016). Unifying distillation and privileged information. *ICLR*. Paper.
- [8] R. G. Krishnan, U. Shalit, and D. Sontag (2017). Structured Inference Networks for Nonlinear State Space Models. *AAAI*. Paper.
- [9] S. Liu, S. James, A. J. Davison, and E. Johns (2022). Auto-Lambda: Disentangling Dynamic Task Relationships. *Transactions on Machine Learning Research*. Paper.
- [10] K. Pomeroy (2016). Ratings methodology update. *KenPom*, October 4. Practitioner methodology note. [Methodology note](#).
- [11] A. Vaswani et al. (2017). Attention Is All You Need. *NeurIPS*, 30. Publisher.
- [12] K. Cho et al. (2014). Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. *EMNLP*, 1724–1734. Publisher.
- [13] P. Shaw, J. Uszkoreit, and A. Vaswani (2018). Self-Attention with Relative Position Representations. *NAACL-HLT*. Publisher.
- [14] J. Lee et al. (2019). Set Transformer: A Framework for Attention-based Permutation-Invariant Neural Networks. *ICML*, 97, 3744–3753. Publisher.
- [15] T. Gneiting and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *JASA*, 102(477), 359–378. Publisher.
- [16] H. Zhao, R. Tachet des Combes, K. Zhang, and G. J. Gordon (2019). On learning invariant representations for domain adaptation. *ICML*, 97, 7523–7532. Publisher.